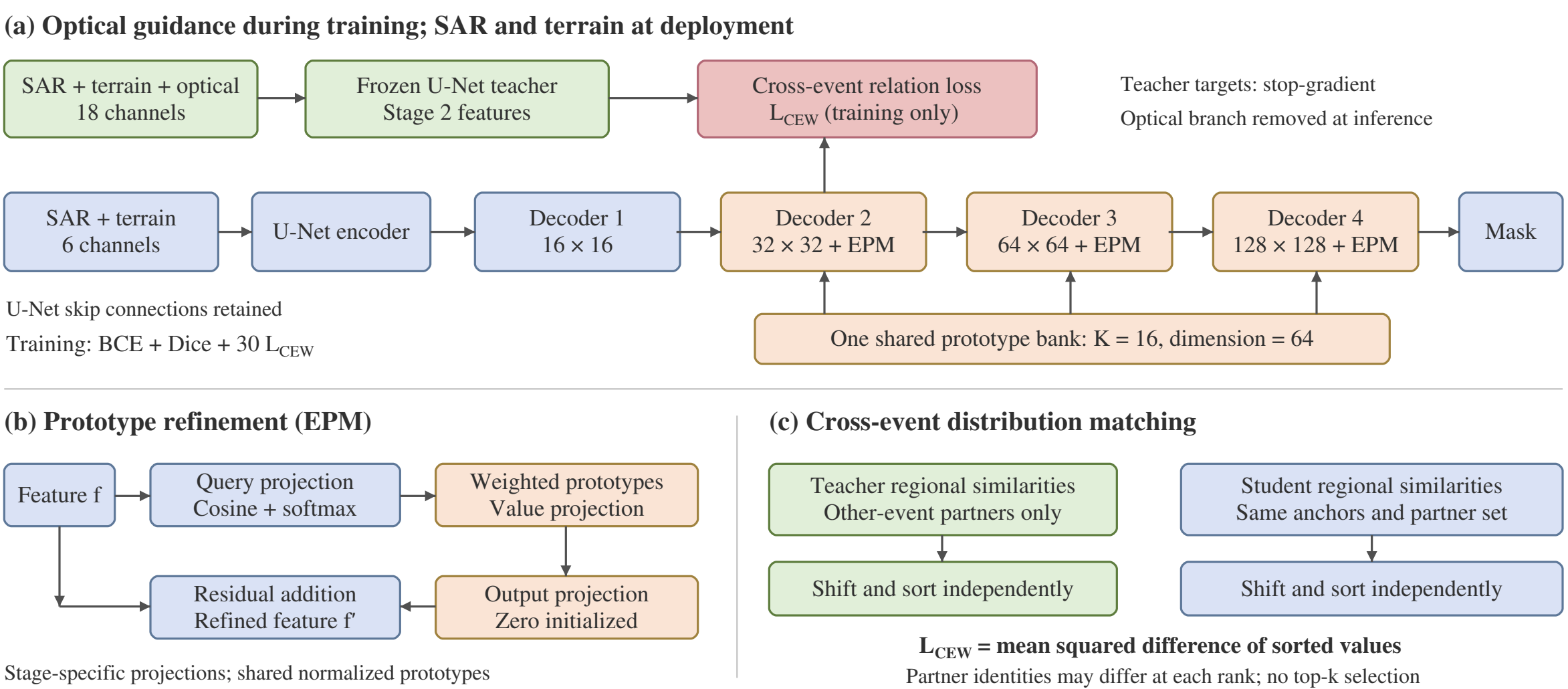

(a) Optical guidance during training; SAR and terrain at deployment
SAR + terrain + optical
18 channels
Frozen U-Net teacher
Stage 2 features
Cross-event relation loss
L_{CEW} (training only)
Teacher targets: stop-gradient
Optical branch removed at inference
SAR + terrain
6 channels
U-Net encoder
Decoder 1
16 × 16
Decoder 2
32 × 32 + EPM
Decoder 3
64 × 64 + EPM
Decoder 4
128 × 128 + EPM
Mask
U-Net skip connections retained
Training: BCE + Dice + 30 L_{CEW}
One shared prototype bank: K = 16, dimension = 64
(b) Prototype refinement (EPM)
Feature f
Query projection
Cosine + softmax
Weighted prototypes
Value projection
Residual addition
Refined feature f′
Output projection
Zero initialized
Stage-specific projections; shared normalized prototypes
(c) Cross-event distribution matching
Teacher regional similarities
Other-event partners only
Shift and sort independently
Student regional similarities
Same anchors and partner set
Shift and sort independently
L_{CEW} = mean squared difference of sorted values
Partner identities may differ at each rank; no top-k selection